

Weaknesses of Sboxes based on 3-round Feistel Structures

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Abstract

The Feistel structure is an attractive approach to construct Sboxes in dimensions larger than 4, as it offers a good trade-off between implementation cost and resistance to classical attacks. As a result, it is used in several lightweight ciphers, such as SCREAM. This work shows that Sboxes built using a 3-round Feistel structure can yield ciphers that are vulnerable to several recently proposed weak-key attacks, including affine commutative cryptanalysis, nonlinear invariants, and fixed-key truncated differential attacks resulting from the clustering of plateau characteristics. Although these attacks differ in nature, they have proven effective against the same targets. By providing simple sufficient conditions on the inner functions of the Feistel Sboxes under which these attacks apply, this work offers a partial explanation of the relationships between them.

Key Words: Sboxes · Feistel structures · Commutative cryptanalysis · Nonlinear invariant attack · Fixed-key differential cryptanalysis

1 Introduction

In symmetric cryptography, the nonlinearity of most block ciphers is ensured by the parallel application of several copies of small nonlinear transformations called Sboxes. They are vectorial Boolean functions from $\mathbb{F}_2^s$ to $\mathbb{F}_2^{s'}$ where s and s' are typically lower than ten. Sboxes must satisfy a range of properties to ensure the resistance of the block cipher to usual cryptanalysis techniques. Among them, linear cryptanalysis [Mat94] and differential cryptanalysis [BS91] — introduced more than 30 years ago — play a key role. In order to lower bound the complexity of these attacks, designers look for Sboxes which have low differential uniformity and low linearity. However,

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the number of possible Sboxes ($2^s!$ for invertible ones) makes a complete classification, even up to appropriate notions of equivalence, impossible as soon as $s > 4$. For $s = s' = 4$, the differential uniformity of any invertible Sbox is greater or equal to 4 and its linearity is greater or equal to 8. The functions achieving these bounds are called optimal and have been classified by Leander and Poschmann [LP07]. However, no similar classification exists for $s = s' = 8$. Even the question of the lowest possible differential uniformity for invertible 8-bit Sboxes is still open, as a corollary of the big APN problem [BDMW10]. Instead, designers often use constructions that allow to build and analyse big Sboxes from smaller ones. The Feistel and MISTY [Mat97] constructions are among the most widespread. In [CDL16] Canteaut, Duval and Leurent derived conditions on the inner Sboxes used in Feistel and MISTY structures that ensure that the resulting Sbox has differential uniformity and linearity as low as possible. The Feistel construction is very attracting to build an Sbox in dimension higher than 4 since it provides a good trade-off between implementation cost and security against classical attacks. That is one of the reasons why several proposals for lightweight encryption, including candidates to the NIST competition used Sboxes based on a Feistel structure. SCREAM, iSCREAM [GLS⁺14], SKINNY64, SKINNY128 [BJK⁺16] and CRYPTON v0.5 [Lim98] are examples of such ciphers.

In the meantime, the emergence of many lightweight ciphers has revealed new types of weaknesses. Several of these new attacks are weak-key attacks in the sense that they identify sets of weak keys for which the cipher is extremely weak. These new weaknesses include three types of attacks, corresponding to different properties and different security criteria for the Sbox. First, the attacks based on nonlinear invariants [TLS16, TLS19, Bey18], suggest that Sboxes should not have quadratic invariants. Moreover, Baudrin and his coauthors [BFL⁺23] showed that the existence of an affine function commuting with the Sbox constitutes a structural weakness for the cipher. Finally, Canteaut and Fruchon [CF26] identified fixed-key differentials with high probability in Sboxes admitting so-called *good sets of differences*. Interestingly, these three attacks apply to the same ciphers (for example MIDORI64 [BBI⁺15] and SCREAM) while they exploit different properties of the Sbox. Therefore, the question of possible links between these attacks has been raised by many authors and partially investigated in [CF26]. This work complements these results by providing simple sufficient conditions on Sboxes based on 3-round Feistel structures that make the resulting cipher vulnerable to each of these three attacks. Most notably, this explains why a cipher like SCREAM is vulnerable to these three types of attacks. The implementation used to validate the results presented in this paper is available online¹.

¹https://github.com/mfru974/wcc26_feistel_structures

2 Preliminaries

Let us introduce some notation that will be extensively used in the rest of the paper. First, when x is a vector of $\mathbb{F}_2^n$ and y a vector of $\mathbb{F}_2^m$, we write $x||y$ for the concatenation of x and y viewed as an element of $\mathbb{F}_2^{n+m}$. When $v_1, v_2, \dots, v_k$ are vectors of $\mathbb{F}_2^n$, we denote by $\langle v_1, v_2, \dots, v_k \rangle$ their linear span over $\mathbb{F}_2$.

When u and v are vectors of $\mathbb{F}_2^n$, we denote by $u \cdot v \in \mathbb{F}_2$ their scalar product. When $u \in \mathbb{F}_2^n$ and $f : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$ is a vectorial Boolean function, we denote by $u \cdot f$ the component of f corresponding to u defined by $(u \cdot f)(x) = u \cdot f(x)$ for all x in $\mathbb{F}_2^m$. In this paper we mainly focus on 3-round classical Feistel structures as defined below.

Definition 1. Let f_i for $1 \leq i \leq 3$ be round functions, where $f_1, f_3 : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$ and $f_2 : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$. The 3-round Feistel structure defined by f_1, f_2 and f_3 is a function from $\mathbb{F}_2^{n+m}$ to itself, denoted by $\text{Feistel}(f_1, f_2, f_3)$, and depicted in Figure 1. When $n = m$ we say that the Feistel structure is balanced.

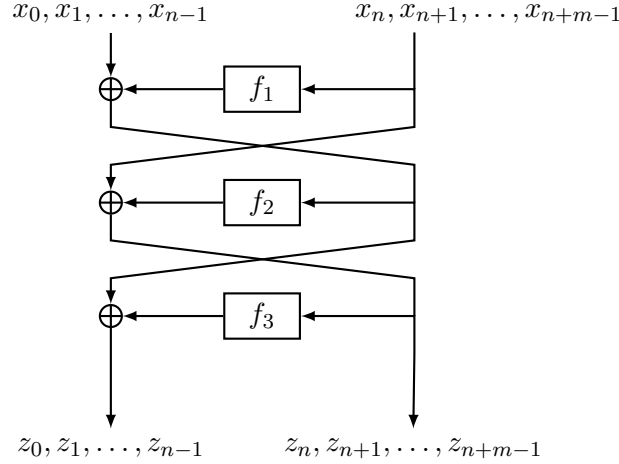


Figure 1: $\text{Feistel}(f_1, f_2, f_3)$ where $f_1, f_3 : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$ and $f_2 : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$

Analysing the resistance of a primitive to differential attacks involves the following classical notion.

Definition 2 (Derivative). For any $S : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^{s'}$ and any $a \in \mathbb{F}_2^s$, the derivative of S at a is the function from $\mathbb{F}_2^s$ to $\mathbb{F}_2^{s'}$ defined by $D_a S : x \mapsto S(x + a) + S(x)$.

3 Commuting with 3-round Feistel structures

Commutative cryptanalysis was first introduced in [Wag04] and then recently revisited in [BFL⁺23]. To attack a block cipher E_k , the core idea is

to find pairs of functions $(\mathcal{B}, \mathcal{C})$ such that $\mathcal{B} \circ E_k = E_k \circ \mathcal{C}$. To simplify the search, the authors of [BFL⁺23] restricted $\mathcal{B}$ and $\mathcal{C}$ to be affine bijections. They found distinguishers on a modified version of MIDORI64 as well as on SCREAM and iSCREAM by using their SPN structures and by analysing each building block separately. Therefore, a key step in this attack is to find a pair of affine bijections (B, C) such that $B \circ S = S \circ C$ where S is the Sbox of the cipher. The authors of [BFL⁺23] provide an efficient algorithm to find all such pairs for a given function S . This algorithm is generic and does not exploit the structure of the Sbox. The following proposition shows that, under some simple condition, 3-round Feistel structures admit non-trivial commuting functions which explains the vulnerability of SCREAM to this attack as identified by Baudrin *et al.* The proof is omitted for brevity.

Proposition 1. *Let us consider $f_1, f_3 : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$ and $f_2 : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$. Let $a \in \mathbb{F}_2^m \setminus \{0\}$, $S := \text{Feistel}(f_1, f_2, f_3)$ and $G_i(x||y) := (x + D_a f_i(y)||y + a)$ for $i \in \{1, 3\}$. Then, $S \circ G_1 = G_3 \circ S$. In particular, when $D_a f_1 = D_a f_3$, $G_1 = G_3$ commutes with S . In addition, if $D_a f_1$ is affine, so is G_1 .*

Application to Scream and iScream

Let us denote by S_1 (respectively S_2) SCREAM's (resp. iSCREAM's) Sbox. These Sboxes are defined by $S_1 = \text{Swap} \circ \text{Feistel}(f_1, f_2, f_3) \circ \text{Swap}$ and $S_2 = \text{Feistel}(f_4, f_4, f_4)$ where Swap denotes the self-inverse function operating on $\mathbb{F}_2^8$ defined by $\text{Swap}(x||y) = (y||x)$ for all $x, y \in (\mathbb{F}_2^4)^2$ and where the f_i s are 4-bit Sboxes whose look-up tables are specified in Table 2 in Appendix.

In the case of SCREAM, the only $a \in \mathbb{F}_2^4 \setminus \{0\}$ such that $D_a f_1 = D_a f_3$ is $a = 0x1$. Moreover, $D_{0x1} f_1$ is affine. According to Prop. 1, this leads to an affine function G_1 that commutes with $\text{Feistel}(f_1, f_2, f_3)$. Since Swap is a self-inverse function, it implies that $\text{Swap} \circ G_1 \circ \text{Swap}$ commutes with S_1 . This is precisely the function found by Baudrin *et al.* in [BFL⁺23].

In the case of iSCREAM, since S_2 uses the same round function three times, Prop. 1 gives $2^4 - 1 = 15$ functions that commute with S_2 . However, since none of the derivatives $D_a f_4$ is affine, neither are the commuting functions. Consequently, commutation with the linear layer and the key addition is not as straightforward to analyse as in the affine case. For this reason, the existence of a non-affine function commuting with the Sbox does not directly lead to an attack. Exploiting such commuting functions therefore remains an open and interesting problem.

4 Nonlinear invariants for Feistel structures

We now focus on a second type of attacks called nonlinear invariant attacks introduced in [TLS16]. Given a block cipher $E_k : \mathbb{F}_2^N \rightarrow \mathbb{F}_2^N$, the general principle is to find a (efficiently computable) Boolean Function $g : \mathbb{F}_2^N \rightarrow \mathbb{F}_2$

such that there exist many weak keys k for which $g(E_k(x)) + g(x)$ is constant over $\mathbb{F}_2^N$. Such a function g is called an invariant for the cipher and, in the particular case where g is affine, it coincides with the notion of invariant subspaces introduced in [LAAZ11, LMR15, GJN⁺16]. To find invariants for a cipher following the SPN construction, the authors of [TLS16] compute invariants for each building-block. In particular, [TLS16, Section 3.2] shows how to extend a quadratic invariant for the Sbox into an invariant for the whole cipher in the case of LS-designs [GLSV15]. Moreover, they provide an efficient algorithm to compute the nonlinear invariants for Sboxes which they use to find the quadratic nonlinear invariants for SCREAM, iSCREAM and MIDORI64. Here, we show that in the case of SCREAM and iSCREAM, the invariants for the Sbox are a direct consequence of the 3-round Feistel structure.

4.1 Invariants over 3-round Feistel structures

In this section, we consider $f_1, f_3 : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$ and $f_2 : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$. The proof of the following proposition is omitted for brevity.

Proposition 2. *Let $S := \text{Feistel}(f_1, f_2, f_3)$ and let u be a vector of $\mathbb{F}_2^n$ such that $u \cdot f_1 = u \cdot f_3 + \varepsilon$ where $\varepsilon \in \mathbb{F}_2$. Then*

$$g(x_0, \dots, x_{n+m-1}) := u \cdot f_1(x_n, \dots, x_{n+m-1}) + u \cdot (x_0, \dots, x_{n-1})$$

is a nonlinear invariant for S .

Application to Scream

As stated in the previous section, SCREAM's Sbox can be written $S_1 = \text{Swap} \circ \text{Feistel}(f_1, f_2, f_3) \circ \text{Swap}$. Moreover, we have $0x2 \cdot f_1 = x_1x_2 + x_0 + x_2$ and $0x2 \cdot f_3 = 0x2 \cdot f_1 + 1$. According to Prop. 2, $g_1(x) := x_5x_6 + x_4 + x_6 + x_1$ is an invariant for $\text{Feistel}(f_1, f_2, f_3)$ and therefore $\tilde{g}_1(x) := x_1x_2 + x_0 + x_2 + x_5$ is an invariant for S_1 . It corresponds to the only quadratic invariant for S_1 found by Todo, Leander and Sasaki in [TLS19].

Application to iScream

Recall that iSCREAM's Sbox can be written $S_2 = \text{Feistel}(f_4, f_4, f_4)$. The space of the quadratic components of f_4 is of dimension 2. More precisely, $0x1 \cdot f_4 = x_0x_1 + x_2$ and $0x4 \cdot f_4 = x_1x_2 + x_1 + x_2 + x_3$ form a basis of that space. Prop. 2 proves that $g_2(x) := x_4x_5 + x_6 + x_0$ and $g_3(x) := x_5x_6 + x_5 + x_6 + x_7 + x_2$ are quadratic invariants for S_2 . However, in [TLS19], the authors exhibit a family of 6 independent non-trivial quadratic invariants for S_2 : g_2 and g_3 are two of them, the other 4 are of different nature. Indeed, they involve monomials of the form x_ix_j with $0 \leq i \leq 3$ and $4 \leq j \leq 7$. Intuitively, since S_2 is a sequence of three identical Feistel rounds, quadratic

invariants for a single Feistel round must lead to quadratic invariants for S_2 . This idea is developed in the following section.

4.2 Invariants for a more general Feistel structure

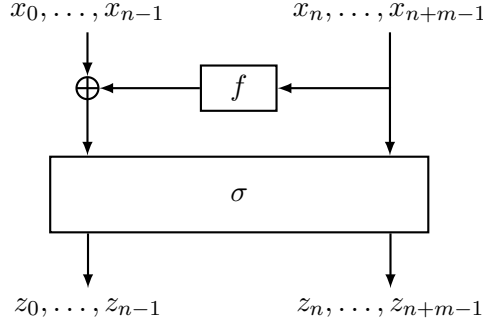


Figure 2: Generalized Feistel round

The classical Feistel structure can be generalized by replacing the swap at the end of each round by any bit permutation σ_i . Fig. 2 represents a round in such a generalized Feistel structure. In the particular case where the round functions of the Feistel structure are all equal, we can derive the following straightforward result.

Proposition 3. *Let us denote by $F : \mathbb{F}_2^{n+m} \rightarrow \mathbb{F}_2^{n+m}$ the result of several iterations of the function depicted by Fig. 2. Let g be such that:*

$$g((x_0, \dots, x_{n-1}) \oplus f(x_n, \dots, x_{n+m-1}) || x_n, \dots, x_{n+m-1}) + g(x) = \varepsilon.$$

and $\forall x \in \mathbb{F}_2^{n+m}, g(\sigma(x)) = g(x)$. Then g is an invariant for F .

Example 1. *The Sbox of SKINNY64 is a 4-round Feistel on $n + m = 4$ bits with $n = 1$, $f : (x_1, x_2, x_3) \mapsto (x_2 + 1)(x_3 + 1)$ and $\sigma(x_0, x_1, x_2, x_3) = (x_3, x_0, x_1, x_2)$. Thanks to Prop. 3, we deduce that $(x_0, x_1, x_2, x_3) \mapsto x_0x_2 + x_1x_3$ is an invariant for this Sbox.*

Moreover, by applying this proposition to iSCREAM's Sbox, one can prove that $g_4(x) := x_0x_4 + x_0x_5 + x_0x_7 + x_1x_4 + x_1x_7 + x_1 + x_2x_6 + x_2x_7 + x_2 + x_3x_4 + x_3x_5 + x_3x_6 + x_3x_7 + x_3 + x_5 + x_6 + x_7$ is another quadratic invariant for S_2 . Note that, among the nontrivial quadratic invariants found by [TLS19], g_4 is the only one such that $g_4 \circ \text{Swap} = g_4$.

5 Differential behaviour of 3-round Feistel structures

In this section we only consider balanced 3-round Feistel structures ($n = m$).

5.1 Previous work on the DDT of 3-round Feistel structures

In differential cryptanalysis, a differential (a, b) describes the propagation of an input difference a to an output difference b through a function. A classical tool for studying the average differential behaviour of an Sbox is the difference distribution table (DDT), which records the probability of each such differential. We now introduce additional notation, to be used throughout this section.

Notation 1. For any $S : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^s$, and any differential (a, b) with $a, b \in \mathbb{F}_2^s$, we denote the input (resp. output) set of (a, b) as follows:

$$\begin{aligned} \text{XDDT}_S(a, b) &:= \{x \in \mathbb{F}_2^s \mid D_a S(x) = b\} \text{ and} \\ \text{YDDT}_S(a, b) &:= \{S(x) \mid x \in \text{XDDT}_S(a, b)\}. \end{aligned}$$

Moreover, $\text{DDT}_S(a, b) := \#\text{XDDT}_S(a, b) = \#\text{YDDT}_S(a, b)$. Finally, when $\text{YDDT}_S(a, b)$ is an affine space, we denote by $\mathbf{V}_S(a, b)$ its underlying vector space.

As mentioned in the introduction, [CDL16] derived conditions on the round functions of 3-round Feistel structures to have the lowest possible differential uniformity. To do so, they expressed some DDT entries of the resulting Sbox as a product of some DDT entries of its inner Sboxes [CDL16, Prop. 4] especially when the input and output differences coincide on their last n coordinates.

Proposition 4 (adapted from [CDL16]). Let f_1, f_2 and f_3 be three n -bit Sboxes with f_2 bijective and $S := \text{Feistel}(f_1, f_2, f_3)$. Then, for all α, α' and β in $\mathbb{F}_2^n$, we have :

$$\text{DDT}_S(\alpha \parallel \beta, \alpha' \parallel \beta) = \text{DDT}_{f_1}(\beta, \alpha) \times \text{DDT}_{f_3}(\beta, \alpha').$$

While the knowledge of the differential uniformity of the Sbox is sufficient to analyse the average differential behaviour of a cipher, additional tools are required to analyse it in the fixed-key setting.

5.2 From good to perfect sets of differences

In [CF26], Canteaut and Fruchon analysed some unexpected fixed-key differential behaviours of several block ciphers. They derived conditions on the linear layer and on the Sbox of a block cipher under which a clustering of plateau characteristics [DR07] occurs. The first condition regarding the Sbox is to admit a *good set of differences*, in the sense of the following definition.

Definition 3 (adapted from [CF26]). A subset $\mathcal{A}$ of elements in $\mathbb{F}_2^s$ is said to be a good set of differences of weight ℓ for $S : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^s$ if all differentials

$(a, b) \in \mathcal{A} \times \mathcal{A}$ are such that $\{(x || S(x)) \mid x \in \text{XDDT}_S(a, b)\}$ is an affine subspace of dimension ℓ of $\mathbb{F}_2^{2s}$ and if there exists a vector space E , called the associated subspace, such that

1. For all $(a, b) \in \mathcal{A} \times \mathcal{A}$, $\text{V}_S(a, b) \subset E$.
2. For all $a \in \mathcal{A}$, $\biguplus_{b \in \mathcal{A}} \text{XDDT}_S(a, b)$ is an additive coset of E .

Moreover, the authors show that SCREAM's and iSCREAM's Sboxes both admit good sets of differences leading to differentials formed by a cluster of plateau characteristics. But SCREAM has a truncated differential of high probability whereas no such phenomenon occurs in iSCREAM. Definition 3 is then refined to capture this distinction.

Definition 4 (Perfect sets of differences). *Let $\mathcal{A}$ be a good set of differences for an Sbox S with associated subspace E . $\mathcal{A}$ is perfect if for all $(\alpha, \beta) \in \mathcal{A}$, there exists $x_S^{\alpha, \beta} \in \text{XDDT}_S(\alpha, \beta)$ such that the following two conditions hold:*

- For all $(a, b, c) \in \mathcal{A}^3$, $x_S^{a,a} + x_S^{b,b} + x_S^{c,c} + x_S^{a+b+c, a+b+c} \in E$.
- There exists $w \in \mathbb{F}_2^s$, such that for all $(a, b) \in \mathcal{A}^2$, $x_S^{a,b} + S(x_S^{b,a}) \in w + E$.

The property of a perfect set of differences is precisely what makes the Sbox weak in the sense that all plateau characteristics within a differential of the cipher share the same weak keys. The difference between the fixed-key differential behaviour of SCREAM and iSCREAM can be attributed to the fact that the good set of SCREAM's Sbox is perfect, whereas the one of iSCREAM is not.

5.3 Good and perfect sets of differences in 3-round Feistel structures

Since the Sboxes of SCREAM and iSCREAM are constructed using a 3-round Feistel structure, the following theorem provides sufficient conditions on the inner Sboxes for the resulting Sbox to admit good (resp. perfect) sets of differences. As a consequence, this allows us to explain the different behaviour of these two closely related ciphers in terms of the properties of the small constituent functions in the 3-round Feistel structures defining their Sboxes.

Theorem 1. *Let $f_1, f_2, f_3 : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ such that f_2 is a bijection. Assume that there exists $\beta \in \mathbb{F}_2^n$ such that:*

- For all $\alpha \in \mathbb{F}_2^n$, $\text{DDT}_{f_1}(\beta, \alpha)$ and $\text{DDT}_{f_3}(\beta, \alpha)$ lie in $\{0, 2\}$
- $\text{Im}(D_\beta f_1) = \text{Im}(D_\beta f_3)$.

Then $\mathcal{A} := \text{Im}(D_\beta f_1) \times \{\beta\}$ is a good set of differences of weight 2 for $\text{Feistel}(f_1, f_2, f_3)$. Moreover, if $D_\beta f_1$ is affine and if there exists $\theta \in \mathbb{F}_2^n$ such that $D_\beta f_1 = D_\beta f_3 + \theta$, then $\mathcal{A}$ is perfect.

According to Prop. 4, if the assumptions of Th. 1 hold, $\text{DDT}_S(\alpha||\beta, \alpha'||\beta)$ can be either 0 or 4. The core idea of the proof of the above theorem is to compute the set $\text{XDDT}_S(\alpha||\beta, \alpha'||\beta)$ in the second case.

Applications to Scream

Recall that SCREAM's Sbox can be written $S_1 = \text{Swap} \circ \text{Feistel}(f_1, f_2, f_3) \circ \text{Swap}$. Using the fact that Swap is a linear involution of $\mathbb{F}_2^8$, one can prove that if $\mathcal{A}$ is an affine good (resp. perfect) set for $\text{Feistel}(f_1, f_2, f_3)$ with associated subspace E , then $\text{Swap}(\mathcal{A})$ is an affine good (resp. perfect) set for S_1 with associated subspace $\text{Swap}(E)$. Consequently, we will focus on the search for good (resp. perfect) sets of differences for $\text{Feistel}(f_1, f_2, f_3)$ using Theorem 1.

It can be checked that f_1 and f_3 are APN and that f_2 is a bijection. Moreover, for all $\beta \in \mathbb{F}_2^4 \setminus \{0\}$, $\text{Im}(D_\beta f_1) = \text{Im}(D_\beta f_3)$, hence S_1 has at least 15 distinct good sets of differences of weight 2 of the form $\{\beta\} \times \text{Im}(D_\beta f_1)$. However, only those corresponding to $\beta \in \{0\mathbf{x}1, 0\mathbf{x}8, 0\mathbf{x}9\}$ are affine spaces and the only one β such that $D_\beta f_1$ is affine is $\beta = 0\mathbf{x}1$. Finally, $D_{0\mathbf{x}1} f_1(x) = D_{0\mathbf{x}1} f_3$. Therefore, Theorem 1 gives a unique perfect set of differences for S_1 . It is an affine subspace of dimension 3 and corresponds to the one which was previously known [BFL⁺23, CF26].

Application to iScream

Recall that iSCREAM's Sbox can be written $S_2 = \text{Feistel}(f_4, f_4, f_4)$. f_4 has no affine derivative. However, for $\beta \in \{0\mathbf{x}3, 0\mathbf{x}7, 0\mathbf{x}9, 0\mathbf{x}b, 0\mathbf{x}f\}$ Theorem 1 shows that $\text{Im}(D_\beta f_4) \times \beta$ is a good set of differences. Among these, the only affine one corresponds to $\beta = 0\mathbf{x}9$ and yields the good set of differences $\mathcal{A}_4$ mentioned in [CF26, Section 6]. The five others were, to the best of our knowledge, previously unknown.

6 Discussion and perspectives

The previous three sections provide sufficient conditions on the round functions f_1, f_2, f_3 of a 3-round Feistel Sbox $\text{Feistel}(f_1, f_2, f_3)$ under which the resulting cipher may be vulnerable to each of the considered attacks. Notably, the second round function f_2 does not influence whether the Sbox admits affine commuting functions or nonlinear invariants. However, this second round function must be a bijection for Theorem 1 to apply. With this in mind, it is instructive to consider the Sbox of LILLIPUT-AE [ABC⁺19].

Indeed, its designers chose to reuse the first and third round functions of SCREAM’s Sbox, while replacing the second one with another bijective function. Our results indicate that this modification does not improve the security of the Sbox against the attacks studied in this paper.

To further illustrate and compare the scope of the above discussed attacks, we summarize several examples of 3-round Feistel structures and their respective weaknesses in Table 1. More precisely, the table indicates ‘Yes’ when the Sbox is weak for one of the three attacks in the sense that it satisfies the sufficient conditions that we have exhibited. These examples demonstrate that no clear relation exists among the three sufficient conditions considered in this work.

Table 1: Comparison of the weaknesses against three attacks for the 3-round Feistel Sboxes constructed from the functions f_i defined in Appendix.

Sbox \ Property	Prop. 1 (commutative)	Prop. 2 (nonlinear invariants)	Theorem 1 (fixed-key differentials)
Feistel(f_1, f_2, f_3)	Yes	Yes	Yes
Feistel(f_1, f_1, f_3)	Yes	Yes	No
Feistel(f_4, f_4, f_4)	No	Yes	No
Feistel(f_5, f_2, f_6)	Yes	No	Yes
Feistel(f_5, f_1, f_6)	Yes	No	No
Feistel(f_1, f_4, f_6)	No	No	No

A Specifications of the Sboxes used in the paper

Table 2: Specifications of the 4-bit Sboxes used in the paper, in hexadecimal notation.

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
f_1	0	2	0	b	3	0	0	a	1	e	0	6	a	4	5	2
f_2	0	2	c	7	5	f	d	6	4	e	8	9	3	1	b	a
f_3	2	0	b	0	0	3	a	0	e	1	6	0	4	a	2	5
f_4	0	8	6	d	5	f	7	c	4	e	2	3	9	1	b	a
f_5	c	5	a	2	5	5	7	6	5	b	0	f	4	3	5	3
f_6	3	1	9	0	6	7	8	2	9	6	d	9	9	5	9	e

f_1 , f_2 , f_3 and f_4 are the Feistel round functions of SCREAM and iSCREAM [GLS⁺14]. f_5 and f_6 were generated at random by the author.

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References

- [ABC⁺19] Alexandre Adomnicai, Thierry P. Berger, Christophe Clavier, Julien Francq, Paul Huynh, Virgine Lallemand, Kévin Le Gouguec, Marine Minier, Léo Reynaud, and Gaël Thomas. Lilliput-AE: a new lightweight tweakable block cipher for authenticated encryption with associated data. Submission to the NIST Lightweight Cryptography Standardization Process <https://csrc.nist.gov/Projects/Lightweight-Cryptography>, 2019.
- [BBI⁺15] Subhadeep Banik, Andrey Bogdanov, Takanori Isobe, Kyoji Shibutani, Harunaga Hiwatari, Toru Akishita, and Francesco Regazzoni. Midori: A block cipher for low energy. In Tetsu Iwata and Jung Hee Cheon, editors, *ASIACRYPT 2015, Part II*, volume 9453 of *LNCS*, pages 411–436. Springer, Berlin, Heidelberg, November / December 2015.
- [BDMW10] Keith A. Browning, John F. Dillon, M.T. McQuistan, and Alan J. Wolfe. An APN permutation in dimension six. In *Post-proceedings of the 9-th International Conference on Finite Fields Their Appl.*, volume 518, pages 33–42. American Mathematical Society, 2010.
- [Bey18] Tim Beyne. Block cipher invariants as eigenvectors of correlation matrices. In Thomas Peyrin and Steven Galbraith, editors, *ASIACRYPT 2018, Part I*, volume 11272 of *LNCS*, pages 3–31. Springer, Cham, December 2018.
- [BFL⁺23] Jules Baudrin, Patrick Felke, Gregor Leander, Patrick Neumann, Léo Perrin, and Lukas Stennes. Commutative cryptanalysis made practical. *IACR Trans. Symm. Cryptol.*, 2023(4):299–329, 2023.
- [BJK⁺16] Christof Beierle, Jérémy Jean, Stefan Kölbl, Gregor Leander, Amir Moradi, Thomas Peyrin, Yu Sasaki, Pascal Sasdrich, and Siang Meng Sim. The SKINNY family of block ciphers and its low-latency variant MANTIS. In Matthew Robshaw and Jonathan Katz, editors, *CRYPTO 2016, Part II*, volume 9815

- of *LNCS*, pages 123–153. Springer, Berlin, Heidelberg, August 2016.
- [BS91] Eli Biham and Adi Shamir. Differential cryptanalysis of DES-like cryptosystems. In Alfred J. Menezes and Scott A. Vanstone, editors, *CRYPTO'90*, volume 537 of *LNCS*, pages 2–21. Springer, Berlin, Heidelberg, August 1991.
 - [CDL16] Anne Canteaut, Sébastien Duval, and Gaëtan Leurent. Construction of lightweight S-boxes using Feistel and MISTY structures. In Orr Dunkelman and Liam Keliher, editors, *SAC 2015*, volume 9566 of *LNCS*, pages 373–393. Springer, Cham, August 2016.
 - [CF26] Anne Canteaut and Merlin Fruchon. Understanding unexpected fixed-key differential behaviours: How to avoid major weaknesses in lightweight designs. In Goichiro Hanaoka and Bo-Yin Yang, editors, *Advances in Cryptology – ASIACRYPT 2025*, pages 97–130, Singapore, 2026. Springer Nature Singapore.
 - [DR07] Joan Daemen and Vincent Rijmen. Plateau characteristics. *IET information security*, 1(1):11–17, 2007.
 - [GJN⁺16] Jian Guo, Jérémy Jean, Ivica Nikolic, Kexin Qiao, Yu Sasaki, and Siang Meng Sim. Invariant subspace attack against Midori64 and the resistance criteria for S-box designs. *IACR Trans. Symm. Cryptol.*, 2016(1):33–56, 2016.
 - [GLS⁺14] Vincent Grosso, Gaëtan Leurent, François-Xavier Standaert, Kerem Varici, François Durvaux, Lubos Gaspar, and Stéphanie Kerckhof. SCREAM & iSCREAM: Side-channel resistant authenticated encryption with masking. Submission to the Caesar competition <https://competitions.cr.yp.to/round1/screamv1.pdf>, 2014.
 - [GLSV15] Vincent Grosso, Gaëtan Leurent, François-Xavier Standaert, and Kerem Varici. LS-designs: Bitslice encryption for efficient masked software implementations. In Carlos Cid and Christian Rechberger, editors, *FSE 2014*, volume 8540 of *LNCS*, pages 18–37. Springer, Berlin, Heidelberg, March 2015.
 - [LAAZ11] Gregor Leander, Mohamed Ahmed Abdelraheem, Hoda AlKhzaimi, and Erik Zenner. A cryptanalysis of PRINTcipher: The invariant subspace attack. In Phillip Rogaway, editor, *CRYPTO 2011*, volume 6841 of *LNCS*, pages 206–221. Springer, Berlin, Heidelberg, August 2011.

- [Lim98] Chae Hoon Lim. Crypton: A new 128-bit block cipher-specification and analysis. *Future Systems*, 1998.
- [LMR15] Gregor Leander, Brice Minaud, and Sondre Rønjom. A generic approach to invariant subspace attacks: Cryptanalysis of Robin, iSCREAM and Zorro. In Elisabeth Oswald and Marc Fischlin, editors, *EUROCRYPT 2015, Part I*, volume 9056 of *LNCS*, pages 254–283. Springer, Berlin, Heidelberg, April 2015.
- [LP07] Gregor Leander and Axel Poschmann. On the classification of 4-bit S-boxes. In Claude Carlet and Berk Sunar, editors, *WAIFI 2007*, volume 4547 of *LNCS*, pages 159–176, Berlin, Heidelberg, 2007. Springer.
- [Mat94] Mitsuru Matsui. Linear cryptanalysis method for DES cipher. In Tor Helleseeth, editor, *EUROCRYPT'93*, volume 765 of *LNCS*, pages 386–397. Springer, Berlin, Heidelberg, May 1994.
- [Mat97] Mitsuru Matsui. New block encryption algorithm MISTY. In Eli Biham, editor, *FSE'97*, volume 1267 of *LNCS*, pages 54–68. Springer, Berlin, Heidelberg, January 1997.
- [TLS16] Yosuke Todo, Gregor Leander, and Yu Sasaki. Nonlinear invariant attack - practical attack on full SCREAM, iSCREAM, and Midori64. In Jung Hee Cheon and Tsuyoshi Takagi, editors, *ASIACRYPT 2016, Part II*, volume 10032 of *LNCS*, pages 3–33. Springer, Berlin, Heidelberg, December 2016.
- [TLS19] Yosuke Todo, Gregor Leander, and Yu Sasaki. Nonlinear invariant attack: Practical attack on full SCREAM, iSCREAM, and Midori64. *Journal of Cryptology*, 32(4):1383–1422, October 2019.
- [Wag04] David Wagner. Towards a unifying view of block cipher cryptanalysis. In Bimal K. Roy and Willi Meier, editors, *FSE 2004*, volume 3017 of *LNCS*, pages 16–33. Springer, Berlin, Heidelberg, February 2004.